

Standard calculation procedures and examples of calculation results for 4-quantization-level, 8-neighbor GLCMs. (By H.Nishiyama / Niigata Univ. 2026/06/27, Ver 2.1)

In PyRadiomics, the angle-related symmetrization parameter `weightingNorm` is set to 'None' by default. GLCMs are generated for each angle to calculate feature values, which are ultimately averaged.

<https://github.com/AIM-Harvard/pyradiomics>

https://github.com/auinen/pyradiomics_edu

It has been confirmed that this processing is likewise applicable to "Radiomics.jl," which runs on Julia.

<https://www.radiomicsjl.org/>

<https://github.com/pzaffino/Radiomics.jl>

Final Result		Results near GLCM-8 (simplified version) and differences	
Average of 4 directions			
Joint maximum	0.18112	0.15714	0.0240
Joint average	2.01818	2.01667	0.0015
Joint variance	0.96322	0.96401	-0.0008
Joint entropy	3.53613	3.60614	-0.0700
		values	differences

8-neighborhood GLCM (normalized)			
0.119	0.143	0.062	0.031
0.143	0.157	0.045	0.048
0.062	0.045	0.010	0.017
0.031	0.048	0.017	0.024

angular direction		8-neighbor GLCM in the angular direction (before normalization)			
0 — 180		16	14	5	4
		14	22	3	6
		5	3	4	2
		4	6	2	2
45 — 225		6	16	6	5
		16	12	7	4
		6	7	0	1
		5	4	1	2
90 — 270		14	18	9	2
		18	12	5	7
		9	5	0	1
		2	7	1	2
135 — 315		14	12	6	2
		12	20	4	3
		6	4	0	3
		2	3	3	4

8-neighbor GLCM in the angular direction (after normalization)	
1	0.143 0.125 0.045 0.036
2	0.125 0.196 0.027 0.054
3	0.045 0.027 0.036 0.018
4	0.036 0.054 0.018 0.018
1	0.061 0.163 0.061 0.051
2	0.163 0.122 0.071 0.041
3	0.061 0.071 0.000 0.010
4	0.051 0.041 0.010 0.020
1	0.125 0.161 0.080 0.018
2	0.161 0.107 0.045 0.063
3	0.080 0.045 0.000 0.009
4	0.018 0.063 0.009 0.018
1	0.143 0.122 0.061 0.020
2	0.122 0.204 0.041 0.031
3	0.061 0.041 0.000 0.031
4	0.020 0.031 0.031 0.041

`weightingNorm` = None

Joint maximum	0.19643
Joint average	2.02679
Joint variance	0.97250
Joint entropy	3.57109
Joint maximum	0.16327
Joint average	2.05102
Joint variance	0.96678
Joint entropy	3.57317
Joint maximum	0.16071
Joint average	1.96429
Joint variance	0.94515
Joint entropy	3.48574
Joint maximum	0.20408
Joint average	2.03061
Joint variance	0.96845
Joint entropy	3.51451

`weightingNorm`
= 'no_weighting'

8-neighborhood GLCM (before normalization)			
50	60	26	13
60	66	19	20
26	19	4	7
13	20	7	10